

Connecting to Yellowbrick

Module 4: Connecting to Yellowbrick

Purpose: Connect to Yellowbrick from a variety of tools and establish secure, well-configured sessions.

Main Topics: ODBC · JDBC · ybsql · DBeaver · Connection security · Best practices

Audience: Business users, analysts, and IT — anyone connecting to Yellowbrick

What You Need to Connect

Every connection to Yellowbrick requires exactly four pieces of information.

HOST The hostname or IP of the Yellowbrick system — provided by your DBA.

PORT Almost always 5432 — the default Yellowbrick port.

DATABASE The name of the database you are connecting to (e.g. melco_analytics).

CREDENTIALS Your username and password.

```
Connect: ybsql -h [host] -p 5432 -d melco_analytics -U [username]
```

Connection Options at a Glance

Yellowbrick uses the PostgreSQL wire protocol — most PostgreSQL-compatible tools connect out of the box.

Tool	Type	Best For	Where to Get It
ybsql	Command-line	Quick queries, scripting, admin tasks	Included in the ybtools package. Ask your DBA or download from support.yellowbrick.com → Downloads.
DBeaver	GUI client	Day-to-day querying, data exploration, export	Free open-source tool. Download the Community Edition from dbeaver.io .
ODBC	Driver	Excel, Power BI, SSRS, reporting tools	Download the PostgreSQL ODBC driver (psqlodbc) from support.yellowbrick.com → Downloads.
JDBC	Driver	Tableau, Java tools, ETL pipelines	Download the JDBC driver JAR from support.yellowbrick.com → Downloads. Add to your tool classpath.
Python	psycopg2	Scripts, Jupyter notebooks, automation	<code>pip install psycopg2-binary</code> — works with standard PostgreSQL connection parameters.

ybsql: Your Command-Line Access

Connect & Query

```
-- Basic connection
ybsql -h [host] -p 5432 -d melco_analytics -U msuarez

-- Run a single query
ybsql -h [host] -d melco_analytics -U msuarez \
-c "SELECT COUNT(*) FROM mart.casino_revenue;"
```

Useful Commands

```
-- Inside ybsql:
\l          -- list databases
\dt         -- list tables
\d mart.casino_revenue -- describe
\q         -- quit

-- Run a SQL file
ybsql ... -f my_report.sql
```

Key Points

1

Part of ybtools package

Download ybtools from the support portal or ask your DBA. Installs on Windows, macOS, and Linux.

2

Works just like psql

ybsql is Yellowbrick's built-in command-line client. Standard PostgreSQL psql commands all work.

3

Use -c for queries, -f for scripts

Pass -c "SQL" for a one-liner, or -f filename.sql for a script. Add -W to prompt for your password.

4

Set env vars to save typing

Set YBHOST, YBDATABASE, YBUSER in your shell profile so you don't repeat connection flags each time.

DBeaver: The GUI Client

DBeaver is a free, open-source GUI database client — A recommended tool for day-to-day querying.

Why DBeaver?

- Free and open-source — no license cost
- Works on Windows, macOS, and Linux
- Connects to Yellowbrick as a PostgreSQL database
- SQL editor, schema browser, data viewer, and CSV export
- Recommended for analysts — no command line required

Setting Up a Connection

```
1. Download Community Edition
   dbeaver.io -> Download -> Community

2. New Database Connection
   Database -> New Connection -> PostgreSQL

3. Enter connection details:
   Host:      [your-yellowbrick-host]
   Port:      5432
   Database:  melco_analytics
   Username:  [your-username]
```

Tip

In DBeaver: right-click your connection → Edit Connection → PostgreSQL tab → set Schema to your default schema. Saves typing it on every query.

Setting Up DBeaver: Step by Step

Four steps to connect DBeaver to Yellowbrick for the first time.

Step 1 — Download

Go to dbeaver.io and download the Community Edition — it's free.

Available for Windows, macOS, and Linux.

`dbeaver.io`
-> Download -> Community

Step 2 — New Connection

Open DBeaver and click New Database Connection (plug icon in toolbar).

Search for PostgreSQL and select it.

Database -> New Connection
Search: PostgreSQL

Step 3 — Enter Details

Fill in the four required fields:

Host: [your-host]
Port: 5432
Database: melco_analytics
Username: [your-username]

Step 4 — Test & Connect

Click "Test Connection." A green check means you're ready.

Click Finish, then open an SQL Editor to run your first query.

```
SELECT current_user,  
       version();
```

ODBC & JDBC Drivers

ODBC Driver

Connects Windows apps to Yellowbrick

Used by Excel, Power BI, SSRS, and any ODBC-compatible reporting tool.

32-bit and 64-bit versions available

Download the PostgreSQL ODBC driver (psqlodbc) from support.yellowbrick.com → Downloads.

Windows DSN configuration

Create a Data Source Name in Windows: Control Panel → Administrative Tools → ODBC Data Sources. Set Server, Port (5432), Database, Username.

Use sslmode=require

Add sslmode=require to the connection string for encrypted connections. Your DBA will confirm the exact parameters.

JDBC Driver

Connects Java-based tools to Yellowbrick

Used by Tableau, Informatica, Apache Spark, and any JDBC-compatible tool.

Download the driver JAR

Get the PostgreSQL JDBC driver (pgjdbc-ng) from support.yellowbrick.com → Downloads. Add the JAR to your tool's classpath.

JDBC URL format

```
jdbc:postgresql://[host]:5432/melco_analytics?user=[user]&password=[pass]&ssl=true
```

Tableau connection

In Tableau: Connect → More → PostgreSQL. Enter the host, port 5432, database name, and your credentials.

Tip: Both drivers are available from the Yellowbrick support portal: support.yellowbrick.com → Downloads. Ask your DBA for the correct version for your environment.

Demo

YBSQL and DBeaver

Secure Connections (SSL/TLS)

Yellowbrick encrypts connections with SSL/TLS. Here is what you need to know to connect securely.

ENCRYPTION All connections use SSL/TLS by default. Data in transit between your client and Yellowbrick is encrypted.

CERTIFICATE Your client may need to trust the server certificate. Your DBA can provide the CA certificate file if required.

SSL MODE Connection strings accept `sslmode=require` (enforce SSL) or `sslmode=verify-full` (SSL + cert check). Use `require` as a minimum.

PASSWORDS Never hardcode passwords in scripts. Use environment variables or a secrets manager instead.

If you see a certificate warning in DBeaver or ybsql, contact your DBA — do not disable SSL verification in a production environment.

Connection Best Practices

X Leaving sessions open

Idle connections hold memory on the manager node. Yellowbrick has a 2,000-connection limit — runaway clients can block other users.

Fix: Always close connections when finished. Avoid tools that open connections to all databases on startup.

X Hardcoding passwords

Passwords in source code or scripts get exposed in git history, logs, and screenshots — a serious security risk.

Fix: Use environment variables (PGPASSWORD), a .pgpass file, or a secrets manager. Never commit credentials to version control.

! Sharing credentials

Each person should use their own username and password. Shared accounts make auditing impossible and create security risks.

Fix: Request your own account from your DBA. Never share login credentials, even temporarily.

! Too many concurrent connections

Opening many connections at once from a single application (e.g. each thread opens its own connection) wastes resources.

Fix: Use a connection pool (e.g. PgBouncer, or your BI tool's built-in pooling). Limit connections per user with CONNECTION LIMIT.

! Not setting search_path

Omitting the schema prefix in queries relies on search_path being set — this breaks in scripts or when run by another user.

Fix: Set search_path in your connection string or at session start: SET search_path = mart, public;

! Ignoring idle session timeouts

Sessions left idle too long may be terminated by the server. This can cause unexpected errors mid-session.

Fix: Set idle_session_timeout in your connection or ask your DBA about the site policy. Reconnect if your session drops.

Questions?

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